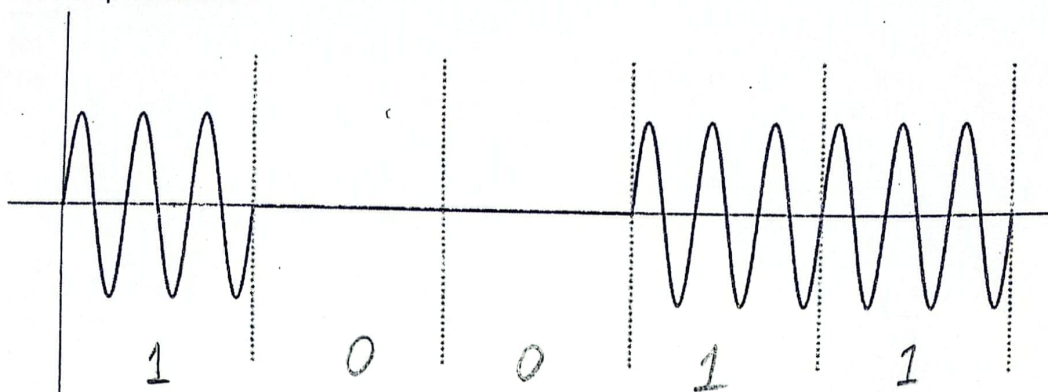


Marks: 20	CSE320: Data Communication	Assignment 3
ID: 22101879	Name: Md Rakib Hossain Ontu	Section: 05

Amplitude Shift Keying (ASK)

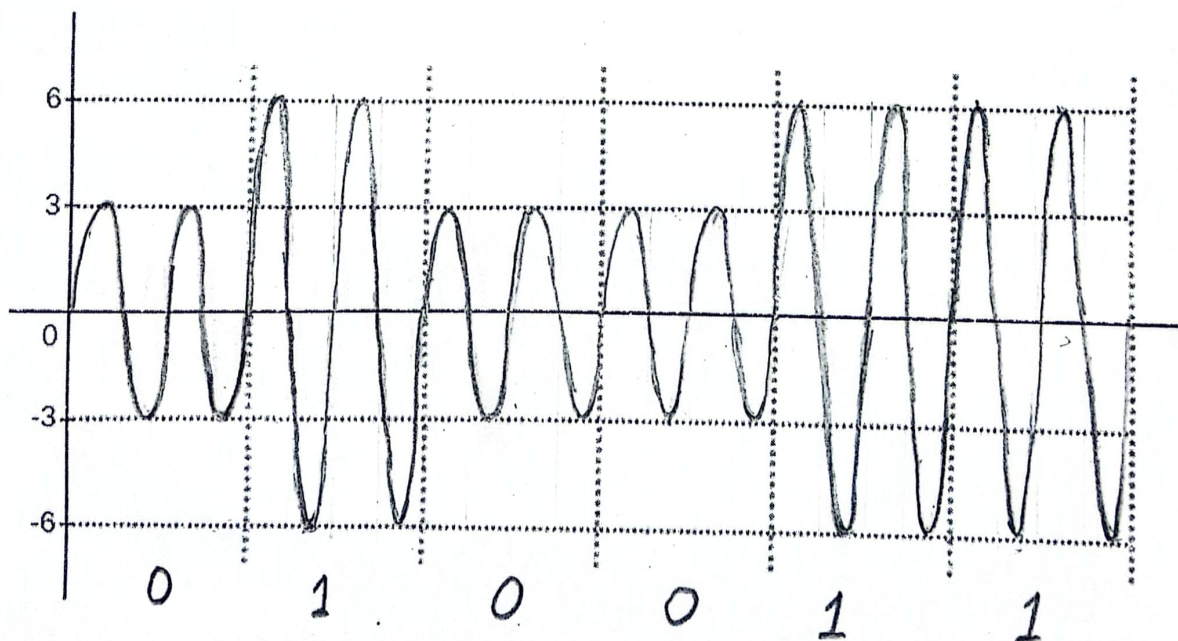
Binary ASK:

1. Determine the digital bit stream from the analog signal below. The signal was modulated using Binary ASK where 0 means signal element with no amplitude and 1 means signal element with amplitude of 3v.



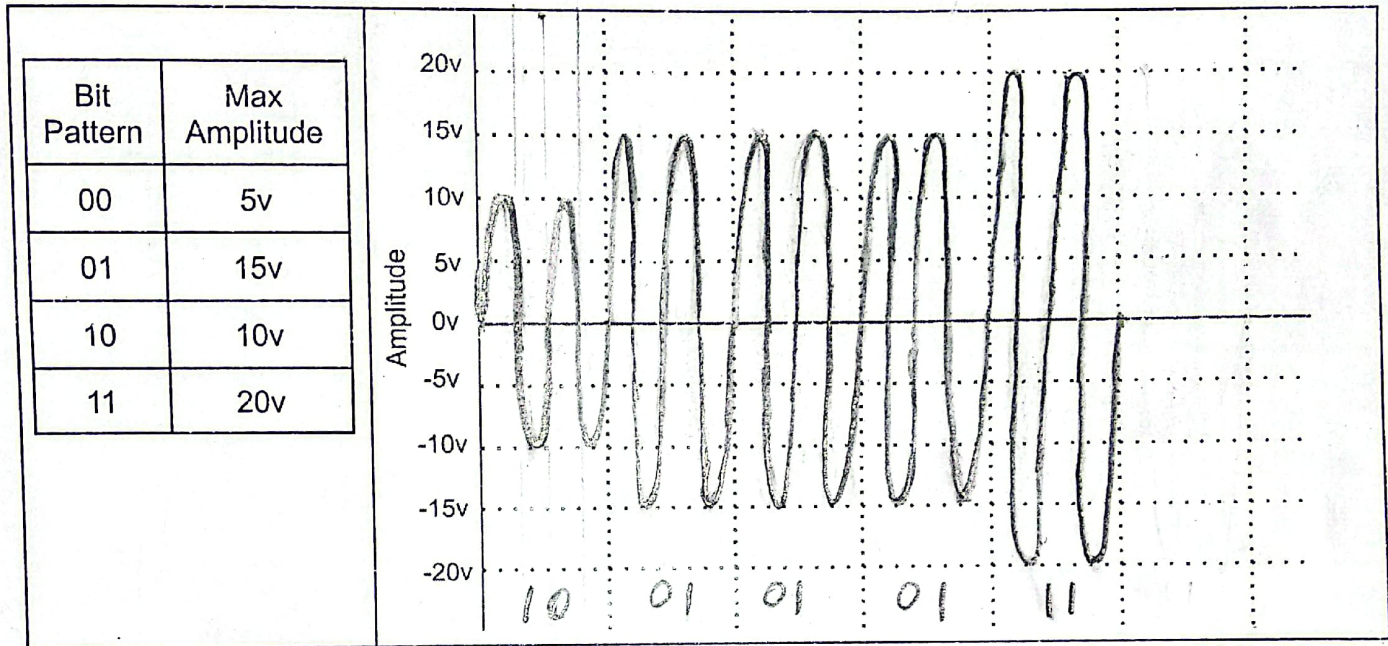
Ans: 10011

2. Draw the analog signal for the digital bit stream 010011 using Binary ASK where 0 means signal element with amplitude of 3v and 1 means signal element with amplitude of 6v. [frequency = 2 for each signal element and phase 0 rad]

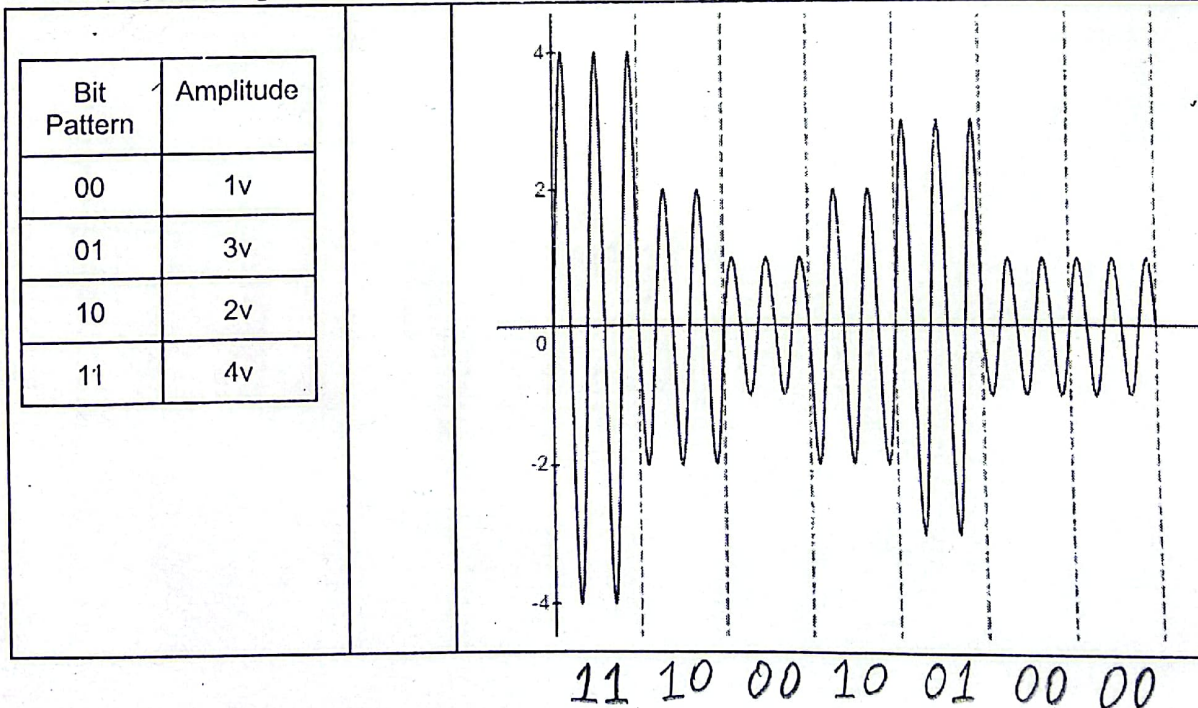


Multi-level ASK

3. In a Multi level ASK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has a frequency of 10 Hz (Each signal element has 2 cycles) and phase is 0 rad. If the amplitude changes according to the following table, draw the modulated signal for the bit sequence 1001010111



4. For the following Multi-level ASK, find the bitstream form the signal below:

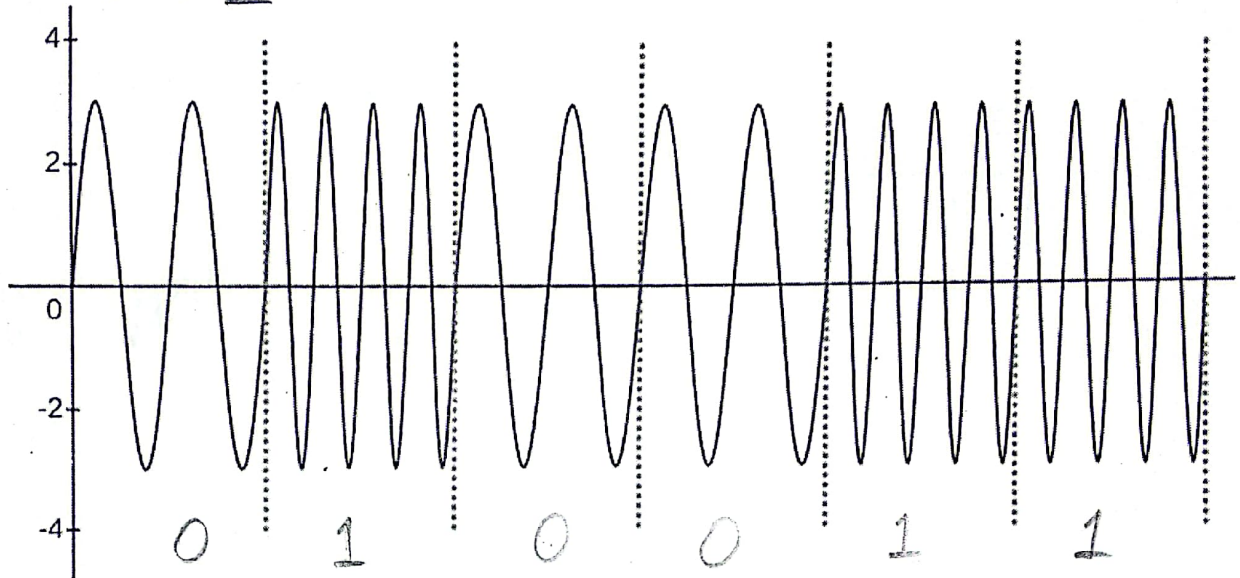


Ans: 111000 10010000

Frequency Shift Keying (FSK)

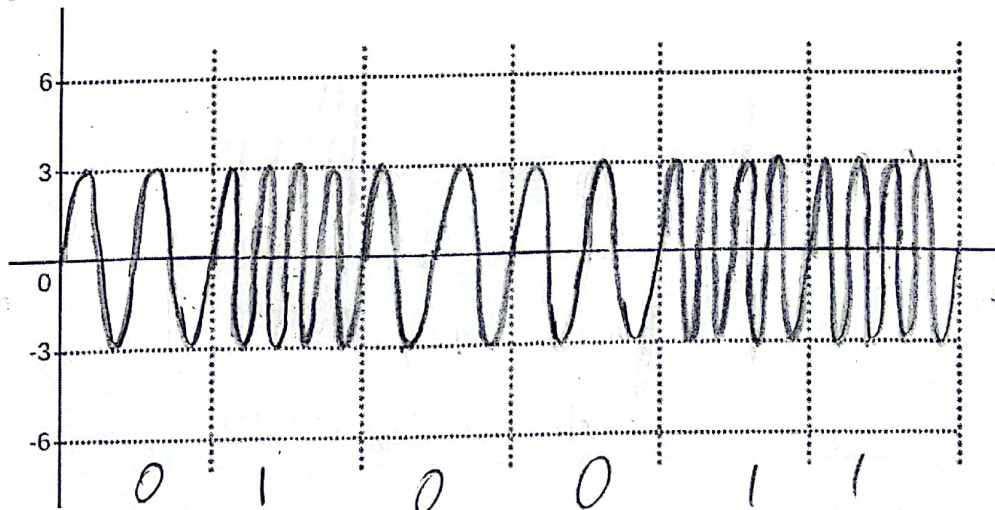
Binary FSK:

5. Determine the digital bit stream from the analog signal below. The signal was modulated using Binary FSK where 0 means signal element with frequency of 2 and 1 means signal element with frequency of 4.



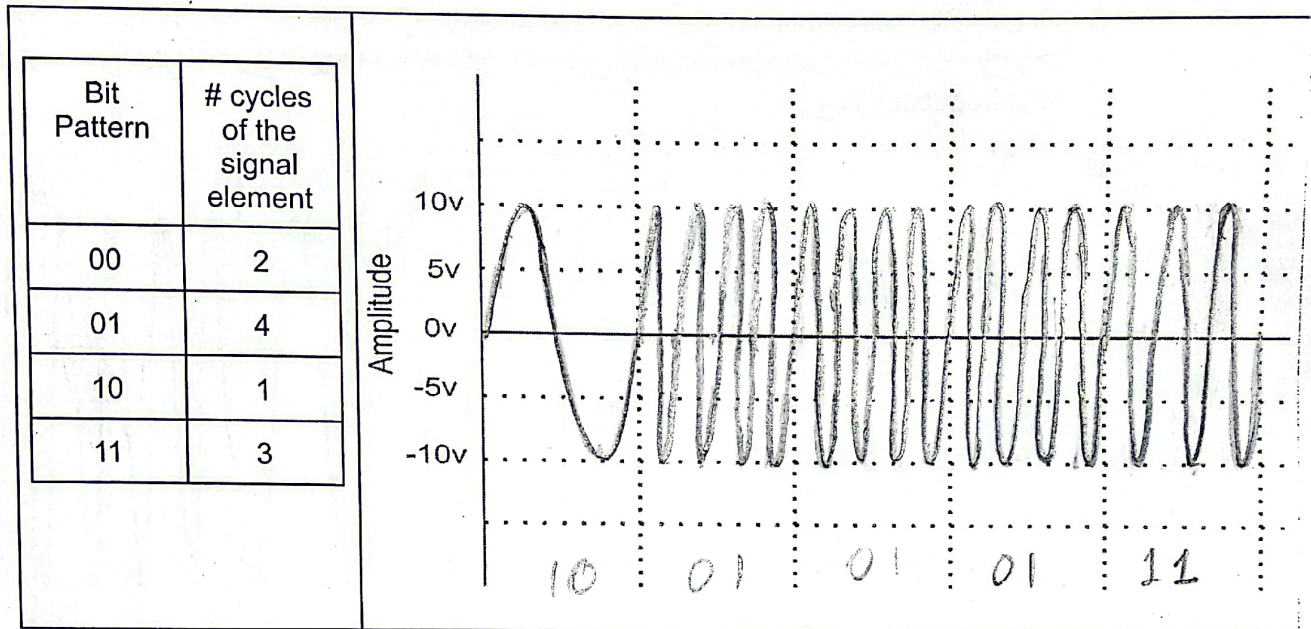
Ans: 010011

6. Draw the analog signal for the digital bit stream 010011 using Binary FSK where 0 means signal element with frequency of 2 and 1 means signal element with frequency of 4.
[Amplitude = 3v and phase 0 rad]

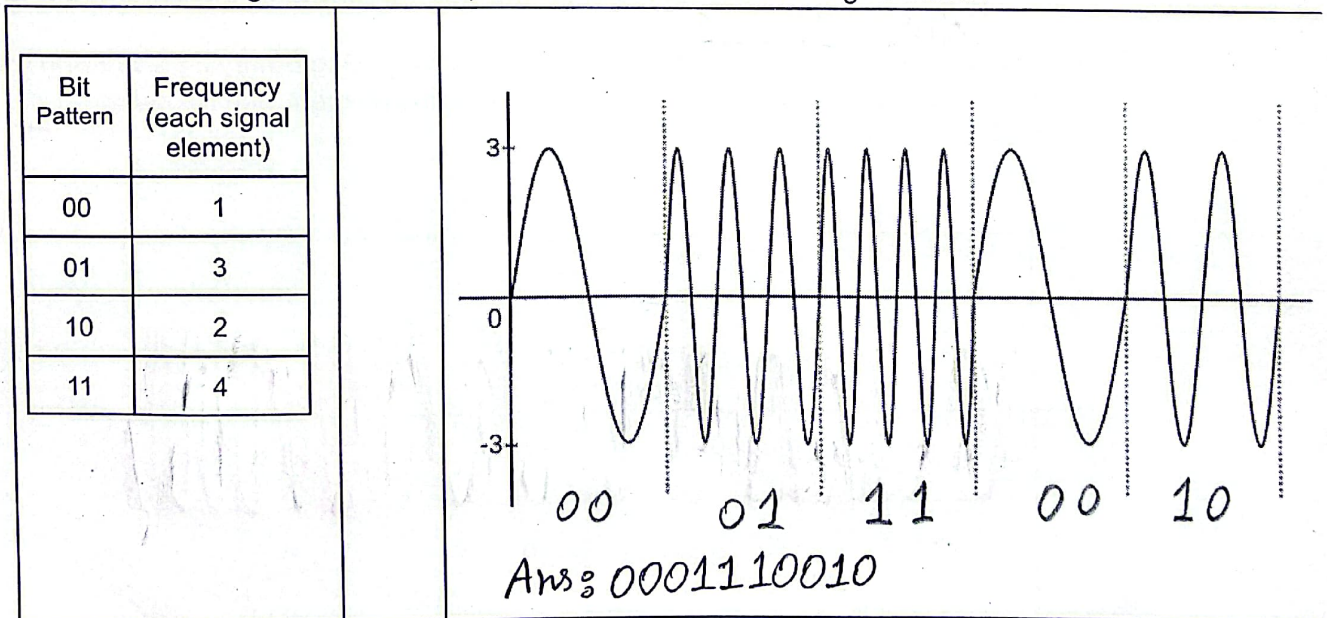


Multi level FSK

7. In a Multi level FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10v and phase is 0 degree. If the frequency changes according to the following table, draw the modulated signal for the bit sequence 1001010111



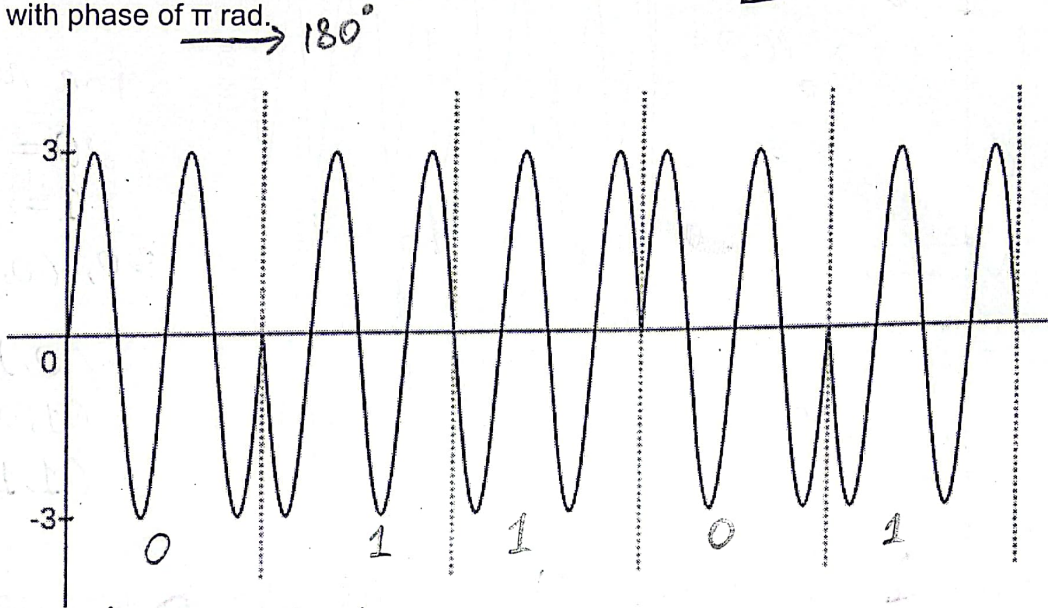
8. For the following Multi-level FSK, find the bitstream form the signal below:



Phase Shift Keying (PSK)

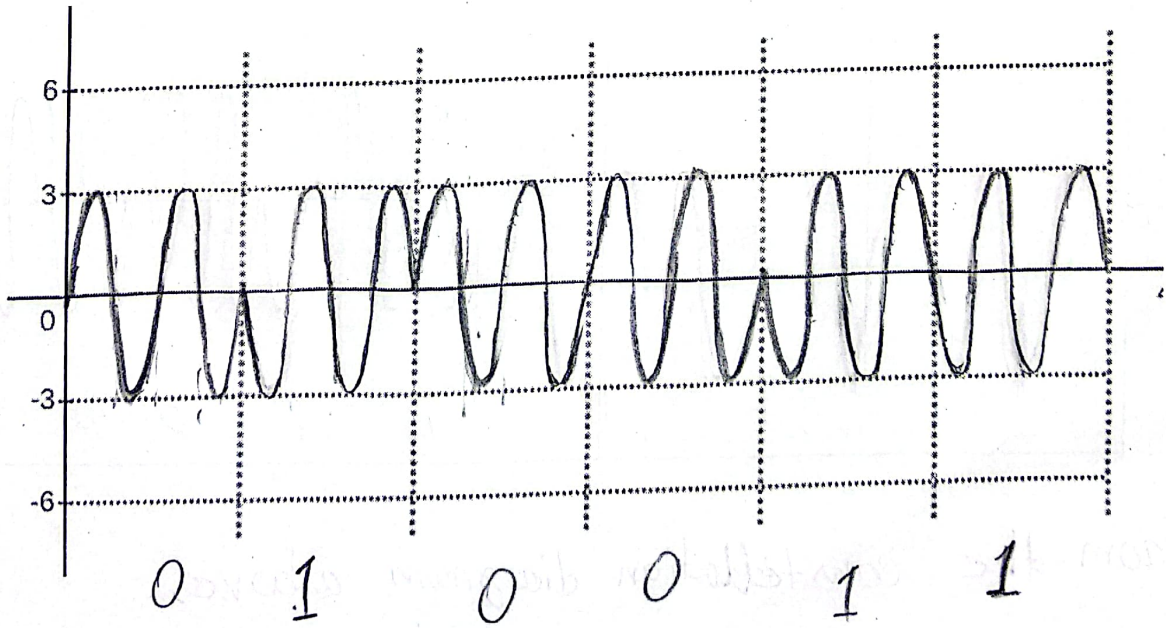
Binary PSK

9. Determine the digital bit stream from the analog signal below. The signal was modulated using Binary PSK where 0 means signal element with phase of 0 rad and 1 means signal element with phase of π rad.



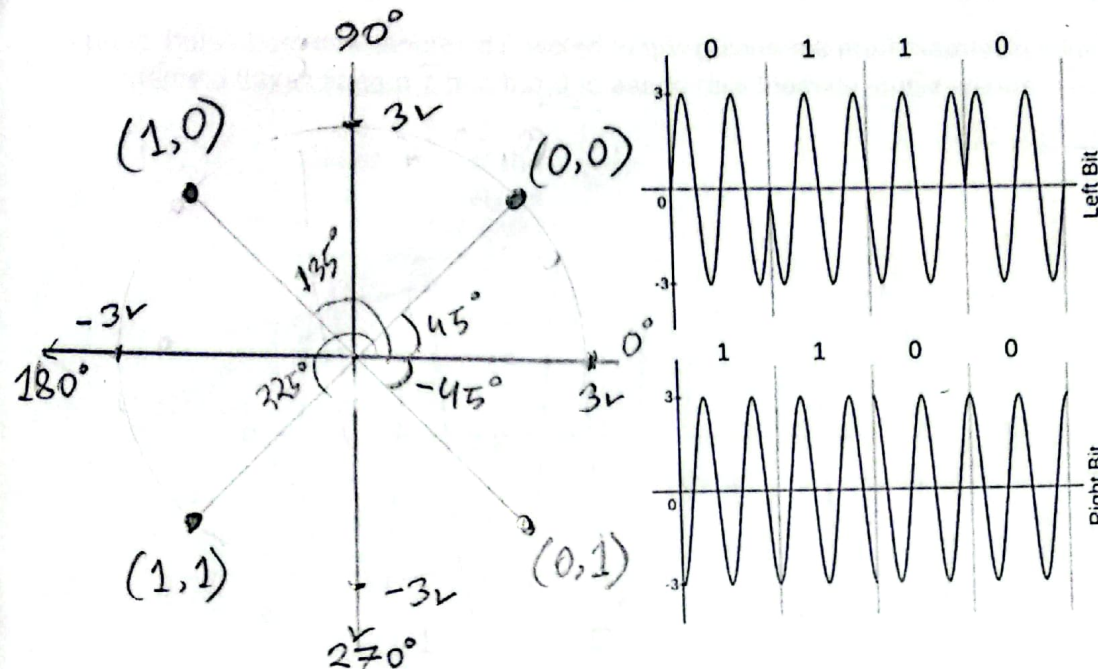
Ans: 01101

10. Draw the analog signal for the digital bit stream 010011 using Binary PSK where 0 means signal element with phase of 0 rad and 1 means signal element with phase of π rad.
[Amplitude = 3V and freq = 2 (for each signal element)]



QPSK & Constellation Diagram:

11. Draw the constellation diagram for the QPSK given below:



From the graphs,
for Left bits,

$$0 = 0^\circ$$

$$1 = 180^\circ$$

for right bits

$$0 = 90^\circ$$

$$1 = 270^\circ$$

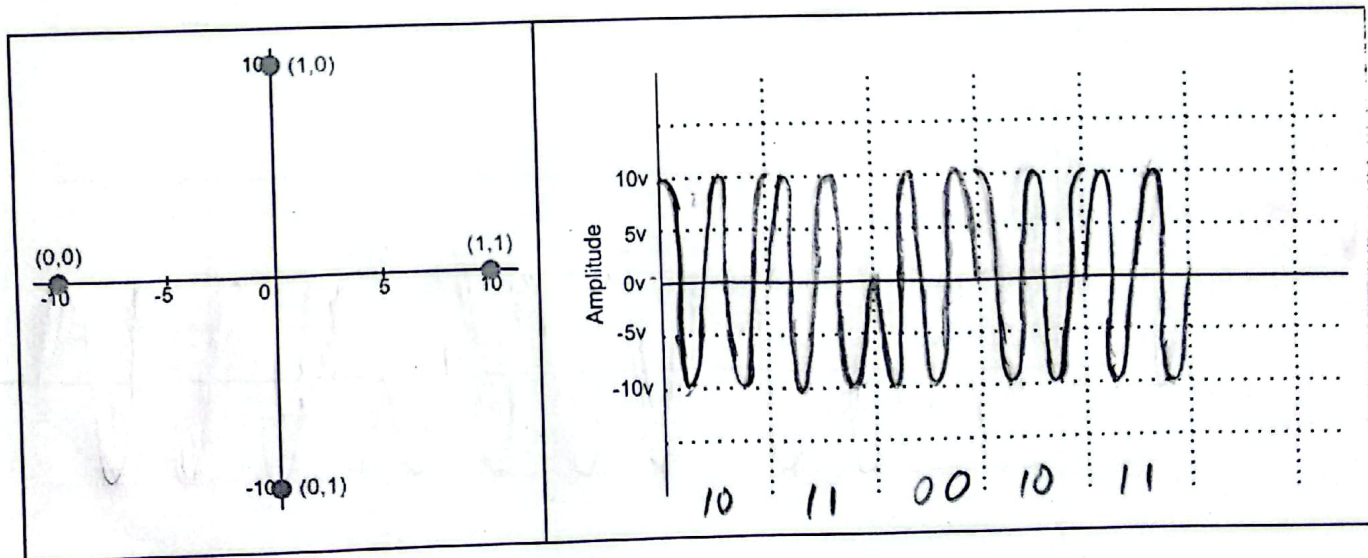
$$\text{So, } (0,0) = \frac{0+90}{2} = 45^\circ$$

$$(0,1) = -45^\circ$$

$$(1,0) = 135^\circ$$

$$(1,1) = 225^\circ$$

12. Draw the analog signal for the bit stream 1011001011 using the constellation diagram given below
[frequency = 2 for each signal element]



From the constellation diagram above,

$$00 = 180^\circ, 01 = 270^\circ, 10 = 90^\circ, 11 = 0^\circ$$

and frequency is ~~10~~ 2 with amplitude of 10V.